W10-10020-LinearRegression

Number of participants: 15

1. **Which of the following best describes the "linearity" assumption in linear regression?**

2 correct answers out of 12 respondents

The independent variables (features) must be linearly independent	25%	3 votes
The predictors (features) must have linear relationships with each other	42%	5 votes
The model must be linear in its parameters	17%	2 votes
The residuals (actual value – predicted value) must follow a straight line	17%	2 votes

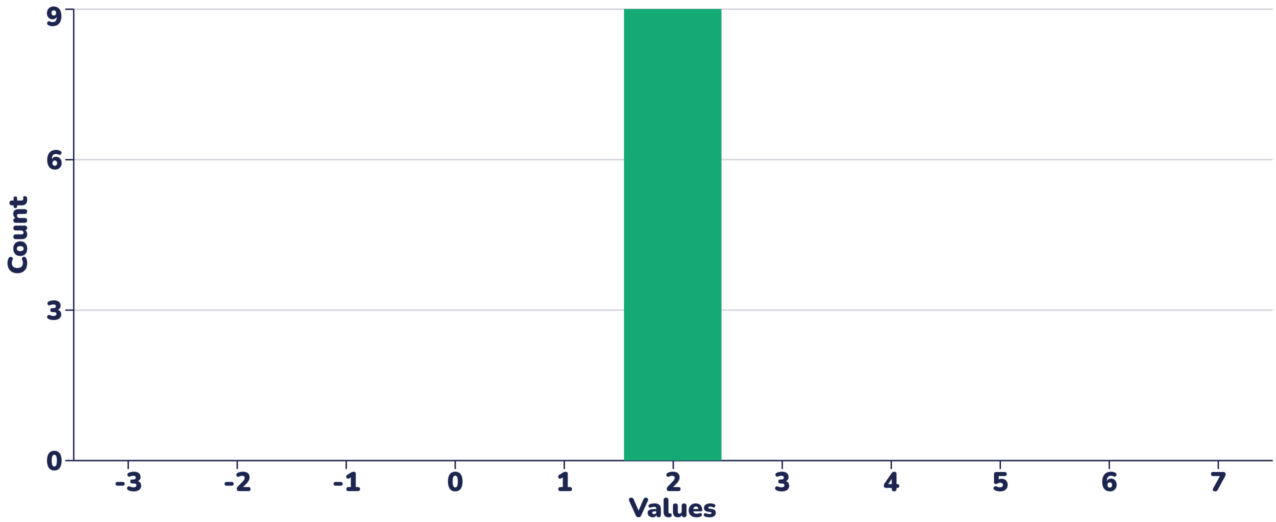
2. You have a linear regression model where predictions are computed as: y_hat = X * beta. X is an n by d matrix (n data points, d features), beta is a d by 1 column vector of weights. Which of the following correctly describes the shape (dimensions) of the predicted output y_hat?

6 correct answers
out of 6 respondents

n by d	0%	0 votes
d by 1	0%	0 votes
✓ n by 1	100%	6 votes
1 by n	0%	0 votes
1 by 1	0%	0 votes

3. You are given the following linear regression model: y = 2.5 + 1.2 * x1 – 0.8 * x2. Use this model to predict y when: x1 = 3, x2 = 5.

9 correct answers
out of 9 respondents



2.1	2.1	2.1	2.1	0	0
Minimum	Mean	Maximum	Median	Standard deviation	Variance

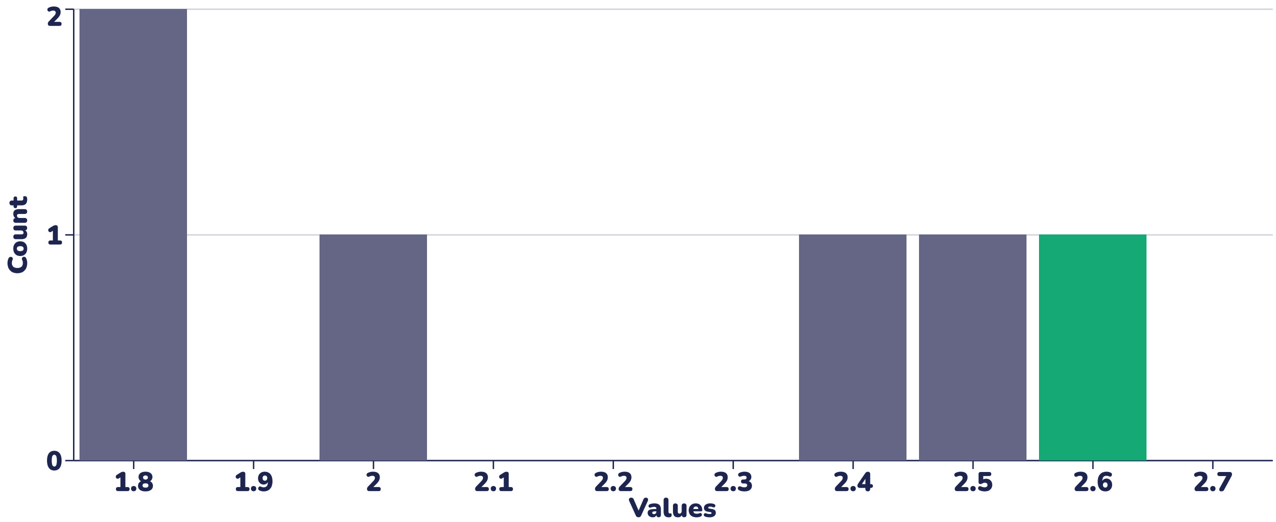
Correct answer

2.1

4. **You are using gradient descent to train the linear regression model: y_pred = a + b1 * x1 + b2 * x2. At the current step, the weights are: a = 2.5, b1 = 1.2, b2 = –0.8. You observe one training example: x1 = 3, x2 = 5, actual y = 4. The learning rate is 0.1. Use mean squared error loss on this one sample (see the screenshot), and compute the updated weights after one step of gradient descent. What are the updated values of a?**

1 correct answer
out of 5 respondents

$$L = \frac{1}{2}(y_{\text{pred}} - y)^2$$



1.8	2.16	2.69	2	0.37	0.14
Minimum	Mean	Maximum	Median	Standard deviation	Variance

Correct answer

2.69

5. **You are using gradient descent to train the linear regression model: y_pred = a + b1 * x1 + b2 * x2. At the current step, the weights are: a = 2.5, b1 = 1.2, b2 = –0.8. You observe one training example: x1 = 3, x2 = 5, actual y = 4. The learning rate is 0.1. Use mean squared error loss on this one sample (see the screenshot), and compute the updated weights after one step of gradient descent. What are the updated values of b2?**

0 correct answer
out of 0 respondent

$$L = \frac{1}{2}(y_{\text{pred}} - y)^2$$

No answers in this question

Correct answer

0.15

6. Which of the following most affects the convergence speed of gradient descent in linear regression?

4 correct answers
out of 5 respondents

Multicollinearity between features (some features are too similar, so the model has trouble telling them apart)	0%	0 votes
The number of categorical variables	20%	1 vote
Whether the features are normalized	0%	0 votes
✓ The choice of learning rate	80%	4 votes

7. You fit two linear regression models on the same dataset: Model A with 2 features and $R2 = 0.85$ and adjusted $R2 = 0.84$. Model B with 10 features and $R2 = 0.89$ and adjusted $R2 = 0.80$. Which of the following is the most reasonable conclusion?

4 correct answers
out of 7 respondents

Model B performs better than Model A in both explanatory power and generalization	0%	0 votes
Model A has worse fit but more overfitting	0%	0 votes
Adjusted R2 always increases with more predictors	43%	3 votes
✓ Model B has higher explanatory power, but may generalize worse than Model A	57%	4 votes

8. You're comparing two regression models on the same dataset: Model X has MAE = 4.2, MSE = 32.1, Model Y has MAE = 3.8, MSE = 51.0. Which of the following is most likely true?

1 correct answer
out of 1 respondent

✓ Model Y has smaller typical errors but more sensitivity to outliers	100%	1 vote
Model Y is better because both error metrics are lower	0%	0 votes
Model X has lower RMSE, hence smaller errors overall	0%	0 votes
The difference in MSE is irrelevant if MAE is similar	0%	0 votes

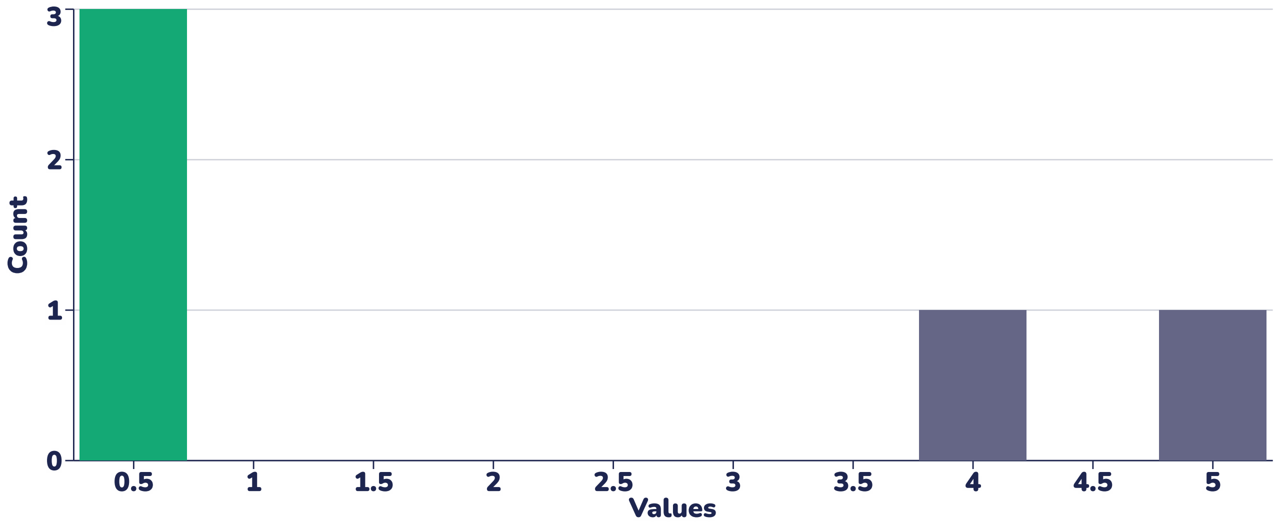
9. A regression model is evaluated on training and test datasets, and it gives: R2 value of 0.96 on training and 0.60 on test set. The MSE is 10.1 on training set and 59.6 on test set. What does this imply?

9 correct answers
out of 9 respondents

The model underfits the training data	0%	0 votes
The model is generalizing well	0%	0 votes
✓ The model is likely overfitting	100%	9 votes
Dataset must be very small	0%	0 votes

10. You are given the following values for a regression model: Total Sum of Squares (SST): 1000, Residual Sum of Squares (SSR): 250. Compute the R2 score.

3 correct answers
out of 5 respondents



0.75	2.25	5	0.75	1.86	3.47
Minimum	Mean	Maximum	Median	Standard deviation	Variance

Correct answer

0.75